DAY 10 | PRE-READ

Never Wait Around

A light read before Day 10. About 15 minutes, no code required.

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Why read this?

Tomorrow you will build a web server in JavaScript using Node.js. Before that, it helps to understand one big idea that makes JavaScript different: it never sits idle waiting for slow things to finish.

1. The coffee shop

Picture two coffee shops. In the first, the barista takes your order, makes the coffee, hands it over, and only then serves the next person. The queue moves slowly. In the second, the barista takes your order, starts the machine, takes the next order, and calls your name when your coffee is ready. Same barista, many more happy customers. JavaScript works like the second shop. This is called **asynchronous** programming.

2. A receipt for something that is coming

When you order, you get a receipt or a buzzer. It is a **promise** that your coffee will arrive. Eventually one of two things happens: you get your coffee, or you are told something went wrong. In JavaScript, a Promise works exactly like that buzzer.

Buzzer status	Promise state
Waiting	pending
Buzzes: coffee ready	fulfilled
Staff apologises: out of milk	rejected

3. Writing it like a recipe

Modern JavaScript has two friendly words, **async** and **await**, that let you write "wait for the coffee, then add sugar" in a natural top-to-bottom style, while the program stays free to serve others.

4. Borrowing tools: npm

Nobody builds everything from scratch. **npm** is a huge free library of ready-made JavaScript tools, like an app store for code. Tomorrow you will borrow a popular tool called **Express** to build a web server in a few lines.

5. Testing without building a screen

How do you test an API before anyone builds the app screens? You use **Postman**, a free tool where you type an address, pick GET or POST, press Send, and see the reply. Please install it before class from postman.com, or use its web version.

Before class: install Postman (desktop or web). VS Code users: also install Node.js LTS from nodejs.org.

6. Think about these before class

- Where else in daily life do you start something, do other things, and come back when it is ready?
- What should an app show you while it is waiting for data from the internet?
- Why is it useful to test an API before the app screens exist?

Kissaten (喫茶店): a traditional Japanese coffee shop